

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
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Section / Batch:	B Tech-AI&DS	Date:	10-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Q1. Q1. E-Commerce Product Sorting System (Merge Sort)

An e-commerce platform must sort millions of products based on price and ratings to ensure fast user access. Analyze how Merge Sort can be used to handle large datasets efficiently. Clearly identify key issues such as stability and scalability. Apply divide-and-conquer concepts to explain the sorting process. Analyze its time complexity and performance. Design a feasible solution and justify why Merge Sort is suitable for this scenario.

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				

Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. E-Commerce Product Sorting System Using Merge Sort

1. Understanding of Case

An e-commerce platform stores **millions of products** with information such as product name, price, rating, and category. To provide users with quick and organized results, the products need to be sorted efficiently.

- The main requirement is to sort products based on **price and customer ratings**.
- Since the dataset is very large, the sorting algorithm must have good performance.
- Simple algorithms such as Bubble Sort and Selection Sort are not suitable because their average and worst-case complexity is **$O(n^2)$** .
- **Merge Sort** is suitable for large datasets because it has a guaranteed **$O(n \log n)$** time complexity.
- Merge Sort is a **stable sorting algorithm**, so products having the same price can maintain their original order.
- The system can use **price as the primary key** and **rating as the secondary key**.
- Merge Sort is also scalable because its performance remains efficient as the number of products increases.

2. Application of Concepts

Merge Sort uses the **Divide and Conquer** technique. The complete product list is divided into smaller lists, sorted, and then combined.

Steps:

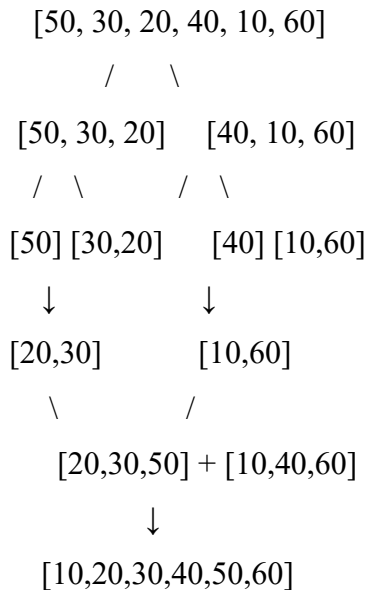
1.Divide: The product list is repeatedly divided into two halves until each part contains one product.

2. Conquer: Each smaller part is sorted recursively.

3. Merge: The sorted parts are compared and combined to form one sorted list.

Example:

Products: [50, 30, 20, 40, 10, 60]



For an e-commerce system, instead of simple numbers, the algorithm compares product records using **price and rating**.

3. Analysis

Time Complexity

Case	Complexity
Best Case	$O(n \log n)$
Average Case	$O(n \log n)$
Worst Case	$O(n \log n)$

Merge Sort divides the dataset into smaller parts, requiring approximately **log n levels**, while merging each level requires **$O(n)$** operations.

Therefore:

$$[T(n) = O(n \log n)]$$

Space Complexity

Merge Sort requires additional memory for the temporary arrays used during merging.

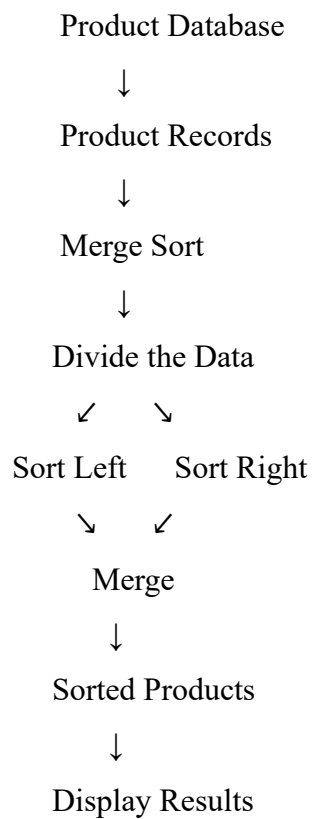
[Space\ Complexity = $O(n)$]

Performance

- Provides consistent performance for large datasets.
- Does not suffer from $O(n^2)$ worst-case performance like simple sorting algorithms.
- Stable sorting helps when multiple products have the same price.
- It is suitable for large databases and external sorting systems.

4. Solution Design

The proposed e-commerce sorting system can work as follows:



Example

Consider four products:

Product	Price	Rating
Laptop A	₹50,000	4.2
Laptop B	₹30,000	4.5
Laptop C	₹50,000	4.7

5. Clarity

The proposed solution clearly explains:

- The e-commerce sorting problem.
- Key issues such as **large data, stability, and scalability**.
- The Divide and Conquer approach.
- Merge Sort's time and space complexity.
- Product sorting based on price and rating.
- The practical system design and its advantages.

The examples and tables make the working of Merge Sort easier to understand.

Conclusion

Merge Sort is an effective solution for an e-commerce product sorting system. Its Divide and Conquer approach allows millions of products to be processed efficiently. With $O(n \log n)$ time complexity, $O(n)$ space complexity, and stable sorting behavior, it provides reliable performance for large-scale product databases. Therefore, Merge Sort is a practical choice for sorting products based on price, ratings, and other attributes.